

Abdullah Munir

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EDUCATION

University of British Columbia

Sep 2022 - Apr 2027

BSc (Honours) in Computer Science (Co-op) - GPA: 4.10/4.33 (Dean's List)

Kelowna, BC

- Received UBC's International Major Entrance Scholarship and Faculty of Science International Student Award
- Relevant Courses:** Object Oriented Programming, Data Structures & Algorithms, Data Analytics, Project Management, Machine Architecture, Artificial Intelligence, Software Engineering, Machine Learning, Parallel Programming (CUDA)

Experience

CGI

Jan 2025 – Apr 2026

Software Engineering Intern

Toronto, ON

- Developed a Gemini-powered **RAG** system to automate Angular UI code generation for CGI's Design System, using semantic chunking and a weighted hybrid search combining **FAISS** and **BM25** to reduce development cycles from weeks to hours.
- Developed AI migration agents to convert **Java RMI** and Swing UI into **Angular** and **Spring Boot** with DAO and MyBatis.
- Architected a migration pipeline on Google Cloud Run, utilizing **GCS FUSE** to mount data for zero-latency agent access.
- Secured migration infrastructure via **Artifact Registry**, automating CVE scanning to prevent false positives in agent images.
- Deployed open-source LLMs on Vertex AI, saving **\$4,000** per month while enabling structured reasoning traces.
- Implemented key agents to automate PostgreSQL, Elasticsearch, and Grafana setups, reducing developer effort by **70%**.

University of British Columbia

Sep 2025 – Apr 2026

Machine Learning Engineer

Kelowna, BC

- Architected a **text detection** pipeline for Cree syllabics using **PyTorch** and **CNN** backbones with confidence scoring.
- Built a custom **data generation pipeline** converting raw image annotations into structured character and word-level CSVs.
- Designed custom evaluation metrics for a **low-resource language**, achieving high detection accuracy on a limited dataset.
- Recreated a weakly supervised two-stage training process to evaluate transferability for Cree syllabic detection.

University of British Columbia, Culture & Technology

May 2024 – Apr 2026

Software Developer

Kelowna, BC

- Developed a scalable Unity-based 3D educational game for **24,000+** students using **C#**, Malbers AI, and modular systems.
- Increased engagement by **30%** through implementing the Observer Pattern to manage state change for time-based quests.
- Architected a **ScriptableObject** system to manage cross-scene state, aligning game-play with learning objectives.

University of British Columbia

May 2024 – Dec 2024

Undergraduate Teaching Assistant & Tutor

Kelowna, BC

- Taught and supported **100+** students in Computer and Data Science through individualized technical guidance and labs.
- Assisted in lectures, labs, and grading for the Computer Creativity course, helping students grasp concepts via office hours.

Projects

AllocAid (TA Allocation System) | TypeScript, Deno, Oak, React

- Developed a containerized Deno **platform** with Role based access control across **4+** user roles and **50+** CRUD endpoints.
- Designed a normalized SQL schema to support a custom scheduling algorithm, reducing manual coordination by **50%**.
- Implemented CI/CD pipelines using GitHub Actions, resulting in a **30%** reduction in deployment time.

TSKFLO (Task-Management System) | React, Redux, Node.js, Socket.IO, JWT

- Architected a full-stack task management system with **15+** CRUD endpoints secured using **JWT**-based role access control.
- Implemented real-time task updates and live messaging with **Socket.IO**, improving responsiveness and team productivity.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, C#, C, SQL

Frameworks & Libraries: React, AngularJS, Node.js, NestJS, Spring Boot, Express.js, React Native, Unity, PyTorch

Tools & Technologies: Git, Docker, GCP, PostgreSQL, MySQL, MongoDB, REST APIs, Jest, Tableau

Concepts: Full-stack Development, Software Engineering, Cloud Services, Machine Learning, Deep Learning, RAG, Fine-tuning, Computer Vision, Automation, Context Engineering, Agile, CI/CD, Microservices